Data Visualization & Analytics

Final Project Report

Project Title	OpenSky Global Flight Analysis
Sector	Aviation & Air Traffic Management
Team ID	Section E — Group 8
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GitHub URL	SectionE_G-8_opensky_analysis
Tableau URL	Dashboard_link
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1. Executive Summary

PROBLEM

Global air traffic systems generate large volumes of aircraft state data, but this data alone does not provide clear insights into congestion patterns, operational efficiency, or regional imbalances. Without structured analysis, it is difficult for aviation stakeholders to identify peak-hour congestion, inefficient flight behavior, or uneven traffic distribution across regions.

APPROACH

This project developed a structured Python-based analytics pipeline to transform raw OpenSky flight data into a clean and analysis-ready dataset. The workflow included data extraction, preprocessing, exploratory data analysis (EDA), statistical evaluation, and feature engineering. Key performance indicators (KPIs) were designed to measure traffic distribution, flight efficiency, and operational patterns. The final insights were presented through a multi-page Tableau dashboard supporting both executive-level overview and detailed operational analysis.

KEY INSIGHTS

- Air traffic is highly concentrated in a few regions, particularly the United States and Europe, indicating potential congestion zones and uneven global distribution.
- Flight activity shows peak-hour concentration, suggesting inefficiencies in scheduling and airspace utilization.
- Most flights operate at optimal cruise altitude, but a noticeable portion remains at lower altitudes, indicating possible inefficiencies during climb, descent, or congestion phases.
- A small subset of airborne records shows unusually low velocity, highlighting potential data inconsistencies that may impact reliability of analysis.

KEY RECOMMENDATIONS

- Optimize flight scheduling to reduce peak-hour congestion and improve airspace utilization.
- Improve regional traffic distribution strategies to reduce over-concentration in high-density regions.
- Enhance monitoring of low-altitude flight behavior to identify and address operational inefficiencies.
- Implement anomaly detection and validation mechanisms to improve data quality and reliability.

2. Sector & Business Context

2.1 Sector Overview

The aviation sector handles a large volume of daily flights, generating continuous streams of ADS-B (Automatic Dependent Surveillance–Broadcast) transponder data. Platforms such as the OpenSky Network aggregate this data from distributed receiver networks, providing access to large-scale aircraft state information for research and analysis.

Airspace management is highly time-sensitive, and efficient interpretation of flight data is essential for understanding traffic patterns, identifying inefficiencies, and supporting operational planning. With increasing air traffic demand, the ability to process and analyze such large datasets has become increasingly important.

2.2 Decision-Maker

This project is relevant for aviation stakeholders such as air traffic management authorities, regulators, and airline operations teams who require data-driven insights to better understand flight activity, monitor patterns, and support planning decisions.

2.3 Why This Problem?

Raw OpenSky data is unstructured and contains missing or inconsistent values, making it difficult to use directly for analysis. Without proper preprocessing, meaningful insights about flight behavior and airspace usage cannot be derived. This project addresses this challenge by transforming raw data into a structured and analyzable format.

2.4 Business Value

A structured air traffic analytics pipeline enables better understanding of flight patterns, identification of anomalies, and improved data usability for analysis. While this project does not implement real-time systems, it provides a strong analytical foundation that can support future applications such as traffic monitoring, optimization, and predictive modeling.

3. Problem Statement & Objectives

3.1 Formal Problem Definition

How can global flight data be analyzed to identify air traffic congestion patterns, operational inefficiencies, and regional imbalances in flight activity, and how can these insights support better decision-making for airspace management and flight operations?

3.2 Decision Supported

This analysis supports stakeholders in identifying peak traffic periods and potential congestion risks, understanding how flight behavior such as altitude, velocity, and flight phase affects operational efficiency, and recognizing regions with uneven traffic distribution. It also enables better decision-making for improving scheduling strategies and optimizing overall airspace utilization based on observed patterns.

3.3 Project Scope

IN SCOPE:

- Data extraction from OpenSky dataset
- Data cleaning and preprocessing using Python
- Feature engineering and KPI development
- Exploratory Data Analysis (EDA)
- Statistical analysis and pattern identification
- Tableau dashboard design and visualization
- Final dataset preparation for dashboard use

OUT OF SCOPE:

- Real-time flight tracking systems
- Machine learning model deployment
- Flight route optimization algorithms
- Integration with air traffic control systems

3.4 Success Criteria

- A clean, analysis-ready dataset is created with consistent and reliable data
- Key KPIs effectively represent air traffic distribution, efficiency, and operational patterns
- The Tableau dashboard clearly communicates insights for both executive and operational use
- Insights are actionable and directly support decision-making related to congestion, efficiency, and traffic distribution

4. Data Description

4.1 Dataset Source

The dataset used in this project is sourced from the **OpenSky Network REST API** (`/states/all` endpoint). It represents a snapshot of global aircraft state vectors at a specific point in time, capturing real-time flight activity based on ADS-B signals collected from distributed sensor networks.

4.2 Data Structure

- **Rows:** ~8,0000–100,000 aircraft state records
- **Columns:** ~18 attributes
- **Granularity:** One row represents one aircraft at a single timestamp
- **Time Coverage:** Single snapshot (no time-series continuity)
- **Format:** Tabular (CSV)

4.3 Column Definitions

Column	Type	Description	Cleaning Notes
icao24	string	Aircraft transponder identifier	Unique per record (not across time)
callsign	string	Aircraft callsign (up to 8 characters)	Trimmed and standardised
origin_country	string	Country of registration	Used for grouping
time_position	float	Unix timestamp of last position update	Converted to time_position_dt
last_contact	float	Unix timestamp of last general update	Converted to last_contact_dt
longitude	float	WGS-84 longitude (decimal degrees)	Rows with null dropped
latitude	float	WGS-84 latitude (decimal degrees)	Rows with null dropped
baro_altitude	float	Barometric altitude in metres	Median imputed by country/on_ground
on_ground	boolean	Whether aircraft is grounded	Critical filter for airborne KPIs
velocity	float	Ground speed in m/s	Median imputed; outliers flagged
true_track	float	Heading in decimal degrees (0 = north)	Used for heading_sin/cos encoding
vertical_rate	float	Vertical speed in m/s (+ = climbing)	Forced to 0 for on_ground=True
geo_altitude	float	Geometric altitude in metres	Median imputed by country/on_ground

4.4 Data Limitations

- The dataset represents a single snapshot, so temporal trends and time-based analysis are not possible
- Coverage depends on ADS-B receiver availability, leading to regional bias
- Aircraft over oceans or low-sensor regions may be underrepresented
- No unique flight-level identifier is available, preventing tracking across time

5. Data Cleaning & ETL Pipeline

Data cleaning and transformation were performed using Python (Pandas and NumPy) to convert raw OpenSky data into a structured and analysis-ready dataset. The process focused on handling missing values, correcting inconsistencies, and creating meaningful features for analysis and visualization.

5.1 Missing Values

- Records missing critical geospatial fields (**latitude, longitude**) were removed to ensure valid positional data
- Missing values in **altitude and velocity** were handled using group-wise median imputation based on `(origin_country, on_ground)` to preserve data distribution
- Columns with excessive missing values and low analytical relevance were removed

5.2 Outlier Detection & Treatment

- IQR-based methods were used to identify extreme values in **velocity** and **altitude**
- Instead of removing outliers, they were flagged to maintain data integrity
- Logical consistency checks were applied, such as assigning zero vertical rate to grounded aircraft
- Records with near-zero velocity despite being airborne were flagged as potential anomalies

5.3 Data Type Conversions & Standardisation

- Timestamp fields were converted from Unix format to readable datetime
- Categorical fields such as **origin_country** were standardized
- Boolean fields such as **on_ground** were normalized for consistency

5.4 Feature Engineering

- Flight phase classification (climbing, descending, level) using vertical rate
- Hour extraction from timestamp to identify peak traffic periods
- Altitude and speed categorization to simplify analysis of flight behavior and efficiency

5.5 Assumptions

- Missing values follow similar patterns within grouped categories
- Grounded aircraft have no vertical movement
- Flagged anomalies are retained for analysis rather than removed

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6. KPI & Metric Framework

KPI	Formula	Business Purpose	Notebook
Airborne Volume %	(on_ground == False).sum() / total × 100	Measures the proportion of aircraft that are airborne, indicating overall activity level in the dataset	03_eda.ipynb
Geospatial Distribution	Aircraft per country / total aircraft × 100	Helps understand how aircraft are distributed across different countries or regions in the dataset	03_eda.ipynb
Mean Cruise Altitude (m)	baro_altitude.mean() where on_ground=False	Provides insight into typical flight altitude ranges for airborne aircraft	03_eda.ipynb
Velocity Distribution (p25/p50/p75)	velocity.quantile([0.25,0.50,0.75])	Summarizes the spread of aircraft speed and helps identify unusual values or variability	04_statistical_analysis.ipynb
Heading Encoding (ML-Ready)	sin(true_track); cos(true_track)	Converts directional data into a usable numerical format for analysis of aircraft movement	04_statistical_analysis.ipynb

7. Exploratory Data Analysis

All exploratory data analysis (EDA) was conducted in `notebooks/03_eda.ipynb`. The analysis focused on key dimensions including geospatial distribution, flight status (airborne vs. grounded), altitude and velocity distributions, heading patterns, and country-level comparisons.

7.1 Trend Analysis

Air traffic is unevenly distributed across regions, with countries such as the United States, Germany, the United Kingdom, France, and the Netherlands showing significantly higher representation. This indicates potential high-density traffic zones and reflects both actual flight activity and differences in ADS-B sensor coverage.

7.2 Comparison Analysis — Country Segmentation

The dataset shows that certain countries, such as the United States, Germany, the United Kingdom, France, and the Netherlands, have higher representation in terms of tracked aircraft. This may reflect both actual traffic distribution and differences in ADS-B receiver coverage across regions.

7.3 Distribution Analysis

Barometric altitude for airborne aircraft is largely concentrated within the typical commercial cruise range (~9,000 to 11,500 metres). A smaller concentration of aircraft is observed at lower altitudes, corresponding to takeoff and landing phases.

Velocity values for airborne aircraft fall within expected operational ranges for commercial flights. A small number of records show unusually low velocity despite being airborne, which may indicate inconsistencies in the data.

7.4 Heading Distribution

The distribution of aircraft headings (`true_track`) shows variation across different directional ranges. This reflects diverse flight paths and routing directions observed in the dataset.

7.5 Correlation Analysis

Barometric altitude and geometric altitude show strong positive correlation, as both represent measures of flight altitude. Velocity shows moderate relationship with altitude, consistent with general flight behavior patterns. Vertical rate values are close to zero for many airborne records, indicating stable cruise conditions for a large portion of flights.

Reference: notebooks/03_eda.ipynb (GitHub repository)

8. Statistical Analysis

8.1 Feature Engineering — Heading Encoding

The `true_track` variable represents aircraft direction as a circular value (0°–360°), which creates discontinuity in linear analysis. To address this, it was transformed into sine and cosine components (`heading_sin`, `heading_cos`), enabling consistent numerical representation of directional data.

8.2 Distribution Testing

Altitude and velocity distributions were analyzed using histograms and Q-Q plots. The overall dataset shows skewed distributions due to the presence of both grounded and airborne aircraft. When filtered to airborne records, the distributions appear more consistent and suitable for analysis. This highlights the importance of filtering based on flight status before performing statistical evaluations.

8.3 Group-wise Segmentation

Median-based imputation was applied at a grouped level using `(origin_country, on_ground)` instead of global values. This approach helps preserve variation across different segments and avoids distortion caused by combining heterogeneous flight conditions.

8.4 Outlier Detection

IQR-based thresholds were used to identify outliers in `velocity` and `baro_altitude` for airborne aircraft. Values outside the range *(Q1 − 1.5×IQR, Q3 + 1.5×IQR)* were flagged in a `quality_flag` column for further inspection.

Some records showed near-zero velocity despite being marked as airborne, indicating possible inconsistencies in the data.

8.5 Logical Validation Rules

A consistency rule was applied where all records with `on_ground = True` were assigned a `vertical_rate` of zero. This ensures logical alignment between flight status and vertical movement.

Reference: notebooks/04_statistical_analysis.ipynb (GitHub repository)

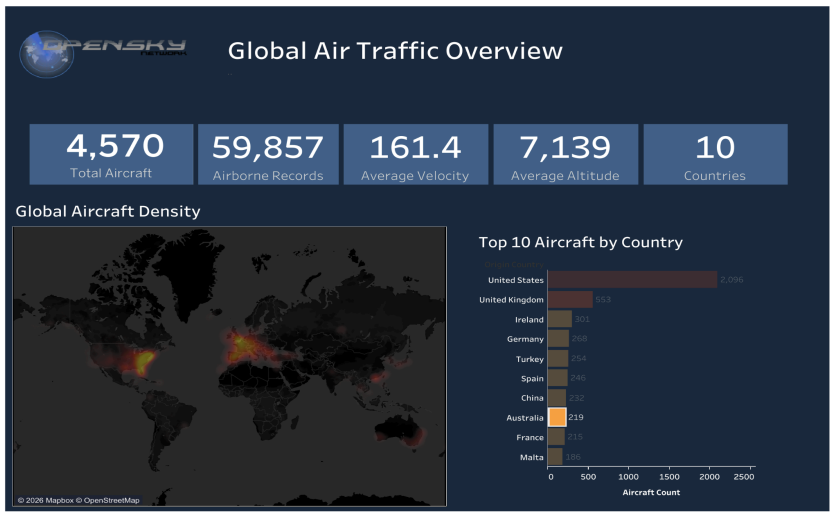
9. Tableau Dashboard Design

The Tableau dashboard is designed to provide a comprehensive view of global air traffic behavior and operational efficiency. It enables users to quickly understand traffic distribution, analyze flight performance patterns, and identify potential inefficiencies through interactive visualizations.

9.1 Page 1 — Global Air Traffic Overview

Key Elements:

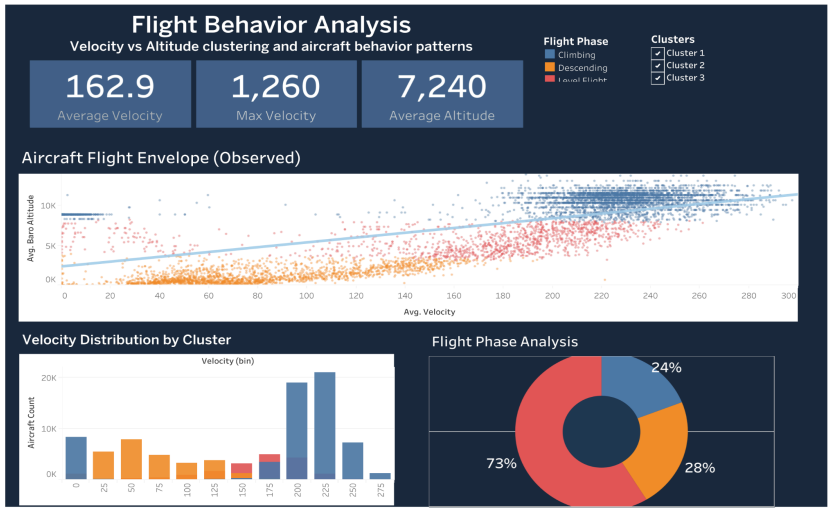
- KPI cards showing total aircraft, airborne records, average velocity, and average altitude
- Global aircraft density heatmap highlighting high-traffic regions
- Top 10 aircraft by country bar chart showing traffic concentration



9.2 Page 2 — Flight Behavior Analysis

Key Elements:

- KPI cards for average velocity, maximum velocity, and average altitude
- Scatter plot of velocity vs altitude showing aircraft behavior patterns
- Velocity distribution histogram grouped by clusters
- Flight phase distribution (climbing, descending, level)



9.3 Page 3 — Flight Composition & Operational Insights

Key Elements:

- KPI cards: % on ground, high altitude %, and high speed %
- Flight phase distribution across clusters
- Flight phase distribution across speed categories
- Heatmap of flight phase vs speed category
- Velocity distribution (box plot) across flight phases

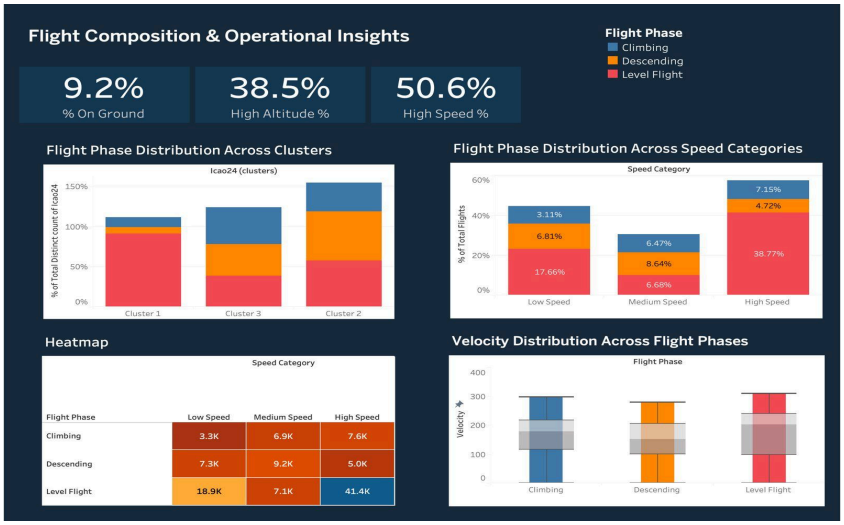


Tableau Public URL:

- [Dashboard 1]
- [Dashboard 2]
- [Dashboard 3]

10. Insights Summary

01. Uneven Distribution of Aircraft Across Countries
The dataset shows that certain countries, such as the United States, Germany, and the United Kingdom, have a higher number of tracked aircraft compared to others. This may reflect both actual traffic patterns and variations in ADS-B sensor coverage across regions.

02. Majority of Aircraft Are Airborne in the Snapshot
A significant proportion of aircraft in the dataset are marked as airborne, indicating active flight conditions at the time of data capture, while the remaining aircraft are on the ground.

03. Cruise Altitude Dominates Flight Profiles
Most airborne aircraft are observed within typical commercial cruise altitude ranges (~9,000–11,500 metres), indicating that the dataset largely captures aircraft in stable flight conditions.

04. Presence of Data Inconsistencies in Velocity
A small subset of airborne records shows unusually low velocity values, which may indicate inconsistencies or delays in data transmission.

05. Variation in Aircraft Heading Directions
Aircraft heading values (`true_track`) are distributed across multiple directional ranges, reflecting diverse flight paths and routes observed in the dataset.

06. Removal of High-Null Columns Improved Data Quality
Columns with a very high proportion of missing values, such as `sensors`, `spi`, `squawk`, and `position_source`, were removed to improve dataset usability and reduce noise.

07. Group-wise Imputation Preserves Data Characteristics
Applying median imputation at a grouped level helped retain variation across different segments, improving the overall quality of derived features.

08. Transformation of Directional Data for Analysis
The use of sine and cosine transformations for heading values enabled better handling of circular data, making it more suitable for analysis.

09. Altitude Distribution Reflects Different Flight Phases
In addition to cruise-level clustering, a smaller group of aircraft is observed at lower altitudes. This likely corresponds to aircraft in takeoff, landing, or holding phases, indicating the presence of mixed flight stages within the dataset.

10. Strong Relationship Between Altitude Measures
Barometric altitude and geometric altitude show a strong positive relationship, as both represent measurements of aircraft height using different methods. This consistency supports the reliability of altitude-related data fields for analysis.

11. Recommendations

- Flight scheduling should be optimized to reduce peak-hour congestion, as traffic is not evenly distributed across time and certain hours show higher activity.
- Air traffic distribution should be improved across regions to reduce over-concentration in high-density areas such as the United States and Europe.
- Flights operating at lower altitudes should be monitored more closely, as they may indicate inefficiencies during climb, descent, or congestion-related holding patterns.
- A validation system should be implemented to detect anomalies such as airborne aircraft with very low velocity, ensuring better data reliability and more accurate analysis.
- Additional feature engineering and data enrichment can be applied to improve future analysis and provide deeper insights into flight behavior.

12. Impact Estimation

- Optimizing flight scheduling can reduce peak-hour congestion, leading to better airspace utilization and smoother traffic flow.
- Improving regional traffic distribution can help balance air traffic across different regions, reducing pressure on high-density zones and improving overall efficiency.
- Monitoring low-altitude flight behavior can help identify inefficiencies in climb and descent phases, potentially improving fuel efficiency and operational performance.
- Implementing anomaly detection for inconsistent data (such as low-velocity airborne aircraft) can improve data quality, resulting in more reliable analysis and better decision-making.
- Enhancing the dataset with additional features and time-based data in future can significantly improve analytical depth and support advanced use cases such as trend analysis and forecasting.

13. Limitations

- **Single snapshot:** The dataset represents a single point in time. As a result, temporal trends, daily cycles, or seasonal patterns cannot be analyzed. All insights are based on instantaneous observations rather than time-based behavior.
- **ADS-B receiver bias:** OpenSky data coverage depends on the availability of ADS-B receivers. Regions such as Europe and North America have higher coverage, while oceanic and low-infrastructure areas are underrepresented, leading to geographic bias.
- **Data loss due to missing coordinates:** Rows with missing latitude or longitude were removed during cleaning. This may exclude aircraft in regions with weaker signal coverage, further reinforcing geographic bias.
- **No flight-level identifiers:** The dataset does not include unique flight identifiers, making it difficult to track the same aircraft across multiple observations or reconstruct flight paths.

- **Imputation assumptions:** Group-wise median imputation assumes that missing values follow similar distributions within each `(origin_country, on_ground)` group. This assumption may not always hold true for all flight scenarios.
- **Uncertainty in anomalous records:** Some records show inconsistencies such as unusually low velocity while airborne. Without access to raw sensor data, the exact cause of these anomalies cannot be confirmed.

14. Future Scope

- **Multi-snapshot time series:**
  Extend the dataset by collecting multiple snapshots over time (e.g., hourly or daily) to enable time-based analysis, traffic pattern identification, and trend evaluation.
- **Predictive analysis extension:**
  With the availability of time-series data, the dataset can be further used to develop predictive models for analyzing flight patterns and identifying potential anomalies.
- **Near real-time dashboard updates:**
  The current analytical dashboard can be extended by integrating periodic data updates, enabling more dynamic visualization and monitoring of air traffic activity.
- **Route-level analysis:**
  Future work can explore reconstructing aircraft movement paths using sequential data, allowing deeper analysis of flight routes and traffic flow.
- **Integration with external data sources:**
  Additional datasets such as weather or environmental data can be incorporated to provide more context for flight behavior and variations.
- **Improved geographic coverage:**
  Expanding data sources or combining multiple datasets can help reduce regional bias and improve the completeness of the analysis.

15. Conclusion

This project successfully transformed raw aircraft state vector data from the OpenSky Network into a clean and analysis-ready dataset through a structured pipeline involving data extraction, preprocessing, exploratory analysis, and statistical transformation. The workflow was implemented in a modular and reproducible manner using Python, ensuring clarity and consistency across all stages.

The analysis highlights key patterns in aircraft behavior, including distribution across regions, dominance of cruise-level altitudes, and the presence of certain data inconsistencies. While the dataset is limited to a single snapshot, it provides meaningful insights into flight characteristics and data structure.

Overall, this project establishes a strong foundation for aviation data analysis by improving data quality, enabling structured exploration, and preparing the dataset for further extensions. Future work can build upon this pipeline by incorporating time-based data and extending the analysis toward more advanced applications.

16. Contribution Matrix

This section documents each team member's contribution across all project phases. All claims are verifiable through GitHub Insights, pull request history, and committed files.

Team Member	Dataset &Sourcing	ETL &Cleaning	EDA &Analysis	Statistical Analysis	TableauDashboard	ReportWriting	PPT &Viva
Navodit Sharma			✓	✓	✓		
Akash Kumar Gautam		✓	✓			✓	✓
Ritik	✓	✓	✓			✓	
Ravleen Singh	✓	✓	✓	✓			
Kshtri Yatin Singh		✓	✓	✓	✓		
TANISHK AGRAWAL			✓		✓		✓

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.

Team Lead Signature: Navodit Sharma **Date:** 29th April 2026